

PROJECT REPORT

Project Name	Aegis-Atlas	Reporting Period
Project Owner	Viren Singh	Nov 30, 2027 to Dec 25, 2027
Prepared by	Viren Singh	

HIGHLIGHTS

- Developed an independent browser-based computational intelligence platform integrating scientific simulations, cyber intelligence concepts, AI-style visualization systems, and interactive STEM exploration tools.
- Designed and implemented multiple interconnected modules including:
 - World Risk Mapping
 - Science Forge simulation systems
 - Cyber Intelligence visualization
 - AI Threat Globe
 - Reports Lab and research documentation tools
- Built over 30 interactive science and engineering models covering:
 - Physics
 - Chemistry
 - Atomic theory
 - Wave optics
 - Collision theory
 - Engineering concepts
- Created browser-native simulation systems without backend infrastructure, focusing on client-side rendering, modular architecture, and interactive visualization techniques.
- Explored AI-inspired system architecture through simulated multi-agent analysis modules, threat propagation systems, and autonomous visualization experiments.

- Developed custom UI systems and data visualization interfaces emphasizing:
 - modular design,
 - responsive interaction,
 - layered information panels,
 - and research-oriented presentation.
- Maintained iterative long-term development with continuous updates, experimentation, and version progression across multiple repositories and project modules.
- Combined interdisciplinary interests across:
 - computational science,
 - simulation systems,
 - visualization engineering,
 - cyber intelligence concepts,
 - and frontend architecture.

CHALLENGES

- Balancing project development with academic responsibilities and maintaining consistent long-term progress while managing school coursework.
- Designing complex browser-based simulations entirely on the client side without access to backend infrastructure or cloud processing systems.
- Structuring multiple independent systems into a unified and coherent platform while maintaining usability and modularity.
- Learning advanced visualization techniques, animation systems, and interactive simulation logic through self-directed experimentation and documentation.
- Optimizing performance for large interactive interfaces containing multiple simulations, animated elements, and layered visualization systems.
- Managing project scope and avoiding excessive feature expansion while keeping the platform understandable and maintainable as a student-led research prototype.
- Creating technically ambitious systems while ensuring the project remained authentic, realistic, and representative of independent student work.

STATUS UPDATES

Task or Deliverable	Task Owner	Status	Notes
Project live	Viren Singh	DONE ▾	The project is live on github see below

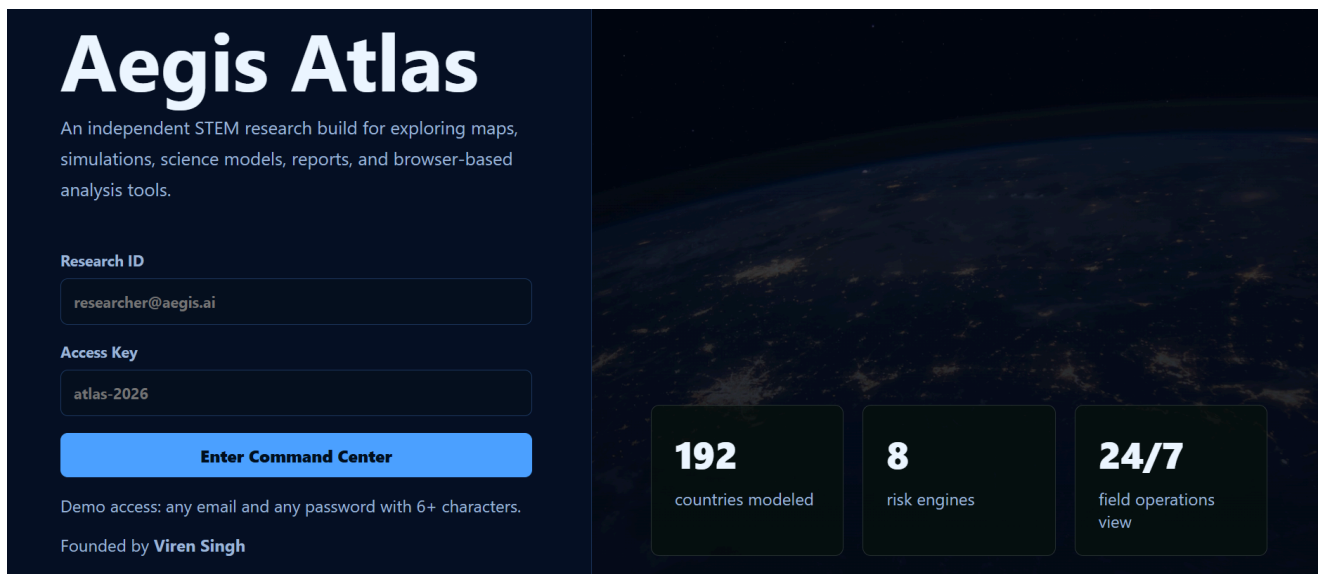
Live project : link

<https://virensingh011.github.io/Aegis-Atlas/>

Github Repository:

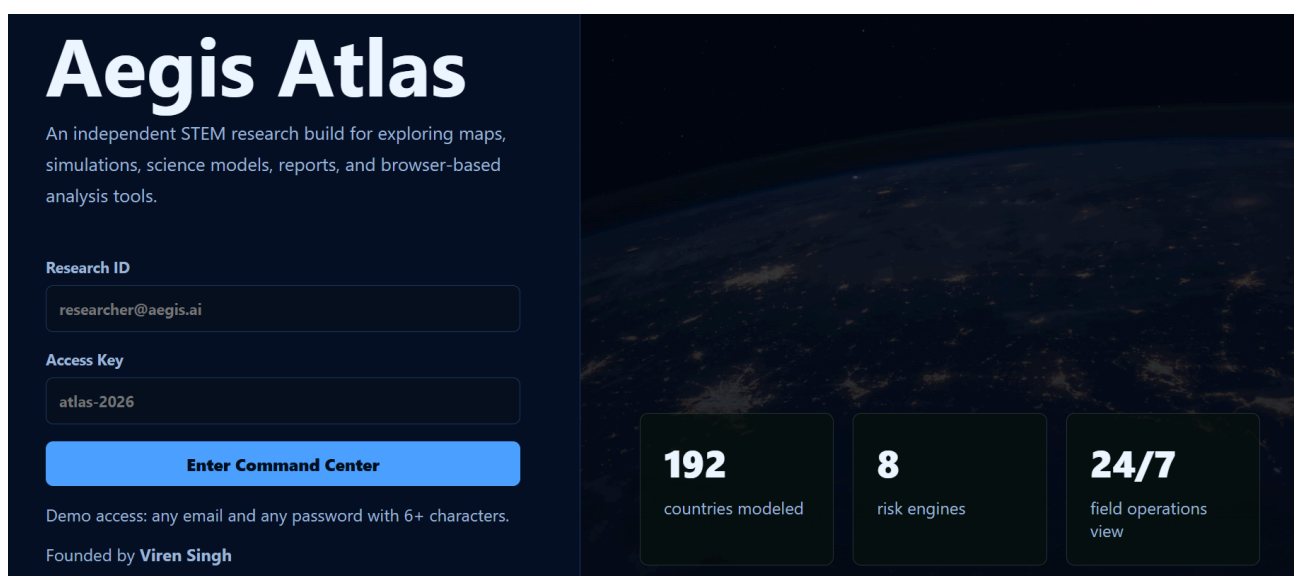
 virensingh011 - Overview

System showcase : screenshots

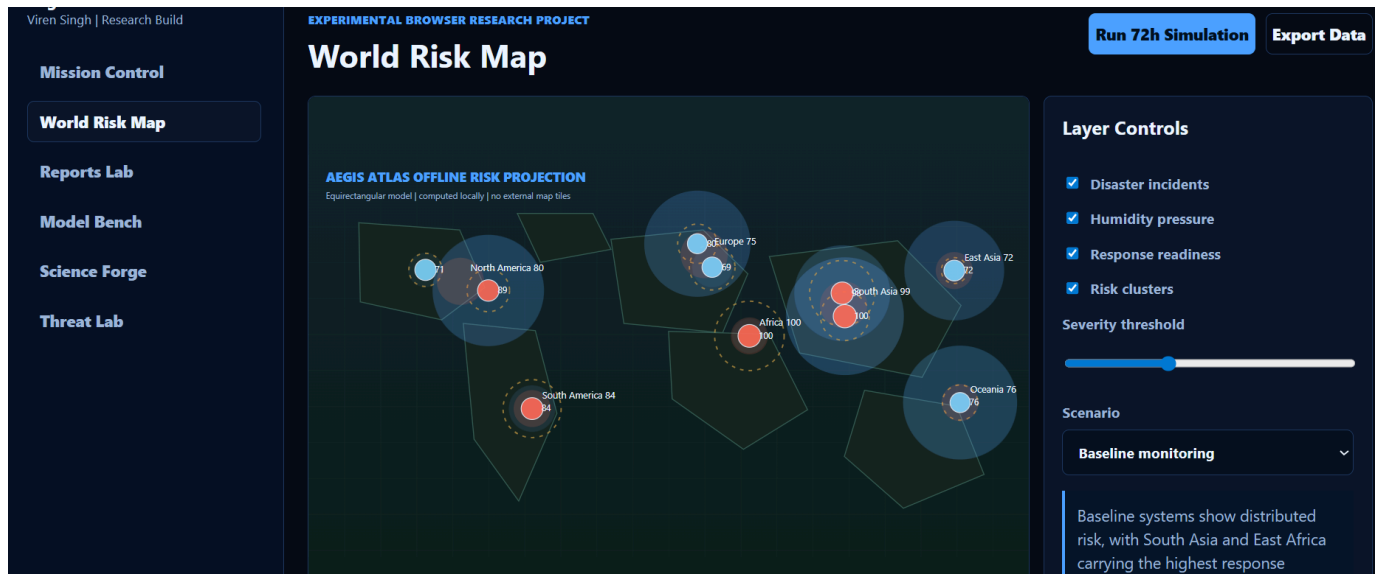


1) The log in system :

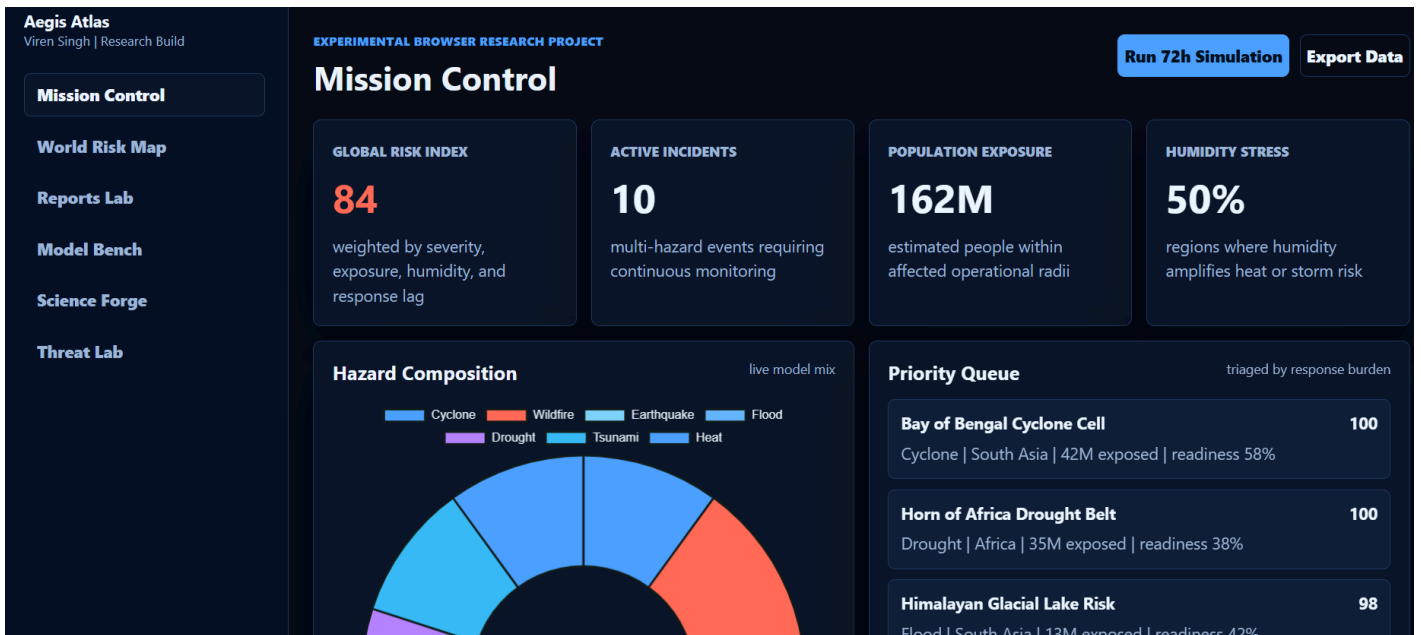
Implemented a secure and responsive login system to manage user access and personalize the platform experience. The authentication flow was designed to support scalability, improve usability, and maintain structured interaction within the simulation environment. Focus was placed on frontend validation, smooth user navigation, and modular system integration during development.



Build in World Risk map



Developed an interactive browser-based global risk visualization system designed to simulate disaster response conditions and regional threat distribution. The platform integrates dynamic risk mapping, layered scenario controls, and configurable simulation parameters to explore how environmental and operational factors influence response readiness across different regions.



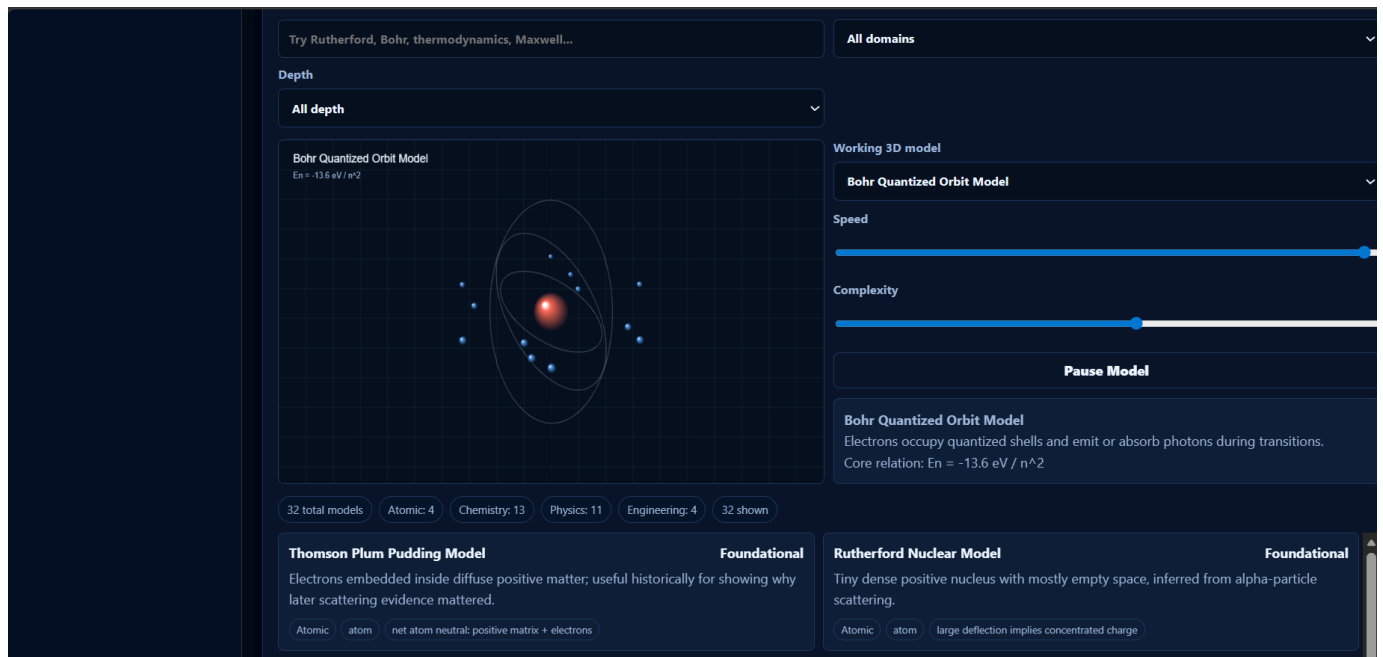
Mission control dashboard

The Mission Control dashboard serves as the central operational interface of the Aegis Atlas research platform. It aggregates real-time simulation metrics, regional hazard exposure, environmental stress indicators, and response-readiness data into a unified monitoring system designed for large-scale disaster analysis.

The system computes a dynamic Global Risk Index using weighted environmental and infrastructure variables, while simultaneously tracking active incidents, projected population

exposure, and hazard composition across multiple disaster categories. A priority queue mechanism identifies high-impact regions requiring urgent attention, enabling structured risk evaluation and scenario-based decision modeling.

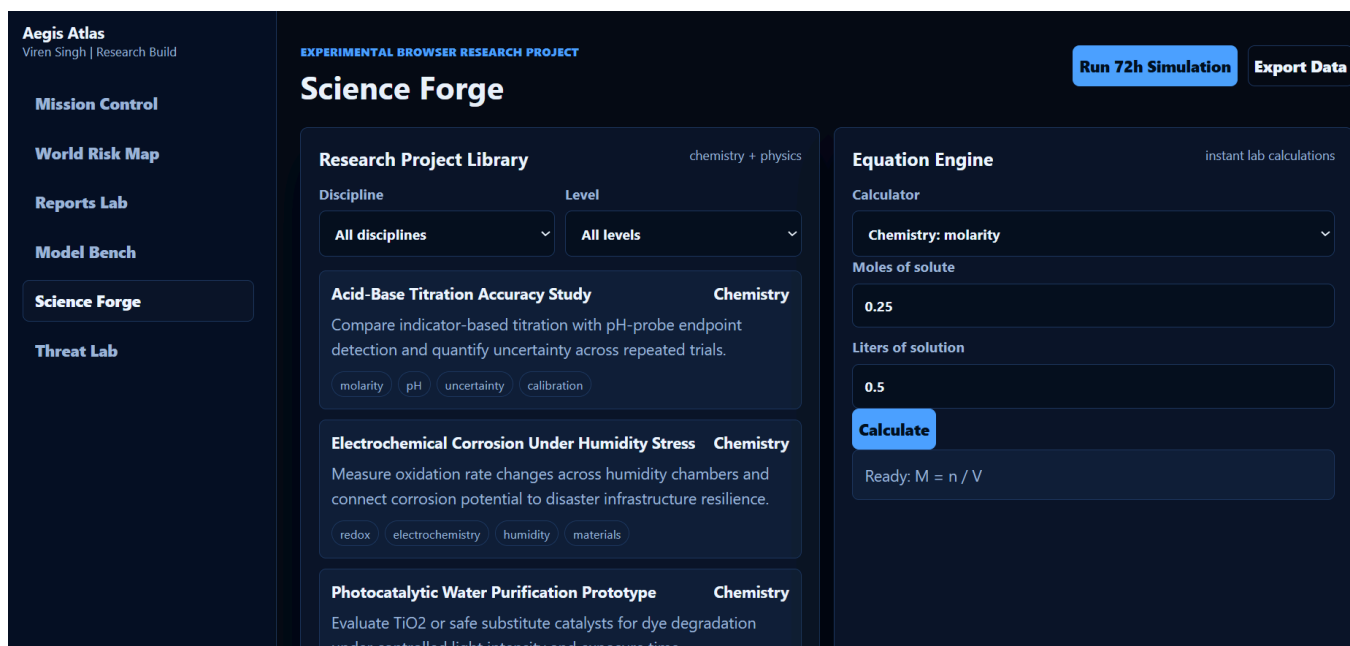
Chemistry and physics 3d model



The laboratory currently includes over 30 interactive scientific models spanning domains such as atomic structure, classical physics, chemistry, and applied engineering. Users can explore historical and modern theoretical systems including the Bohr Atomic Model, Rutherford Scattering Model, Thomson Plum Pudding Theory, thermodynamic relations, and electromagnetic concepts through adjustable simulation parameters and real-time visual rendering.

Each model is paired with contextual scientific explanations, mathematical relations, and adjustable

complexity controls, enabling both conceptual learning and experimental interaction. The system was designed to demonstrate the application of computational visualization, scientific abstraction, and frontend simulation architecture within a scalable research-oriented platform.



Science Forge Research Environment

Science Forge is a browser-based scientific experimentation and computational research workspace designed to integrate chemistry and physics concepts into a unified interactive platform. The system combines virtual research modules, analytical project libraries, and equation-based computational tools to support exploratory scientific learning and applied simulation.

The platform includes a structured Research Project Library containing experimental concepts across chemistry, physics, and environmental science

domains. Each project module presents theoretical objectives, experimental focus areas, and scientific tags to organize interdisciplinary investigation and model-driven analysis.

Aegis Atlas

Viren Singh | Research Build

Mission Control

World Risk Map

Reports Lab

Model Bench

Science Forge

Threat Lab

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Principal Investigator

EXPERIMENTAL BROWSER RESEARCH PROJECT

Run 72h SimulationExport Data

Reports Lab

Report Builderfield-ready synthesis

Region focus

North America

Report type

Scientific assessment

Generate Report

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Generated Intelligence Report12:51:25 PM

AFGIS ATLAS SCIENTIFIC REPORT

Founder: Viren Singh

Scope: North America

Generated: 5/10/2026, 12:51:25 PM

1. Executive Signal

North America currently contains 2 tracked incident models with an average risk score of 80. The leading concern is Gulf Coast Storm Surge, a flood scenario with 16M estimated population exposure and 86% humidity pressure.

2. Highest Priority Incidents

1. Gulf Coast Storm Surge: score 89, severity 7.9/10, readiness 64%, trend +9.

2. California Wildfire corridor: score 71, severity 7.7/10, readiness 73%, trend +8.

3. Research Interpretation

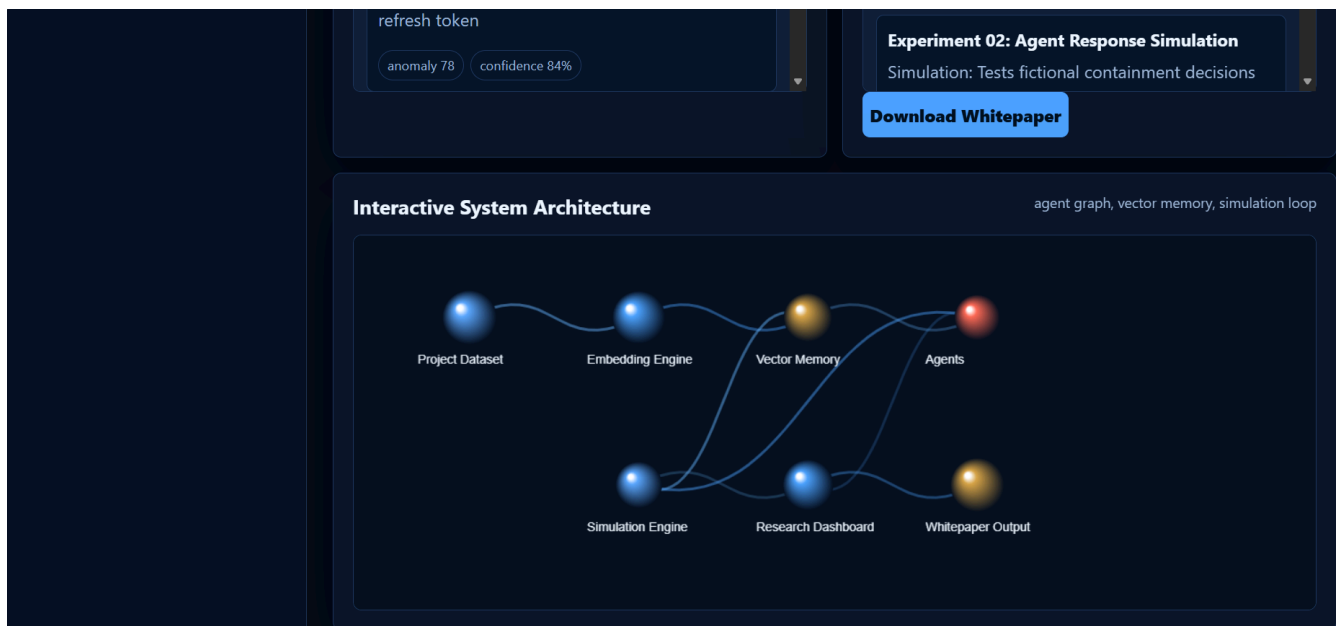
The model weights hazard severity, population exposure, humidity amplification, readiness gaps, and short-term trend acceleration. This is a decision-support simulation designed for research, planning, education, and Github deployment, not a replacement for official emergency alerts.

4. Recommended Action

Prioritize field verification, shelter and hospital capacity checks, public communication, satellite/radar validation, and logistics routing for incidents scoring above 75.

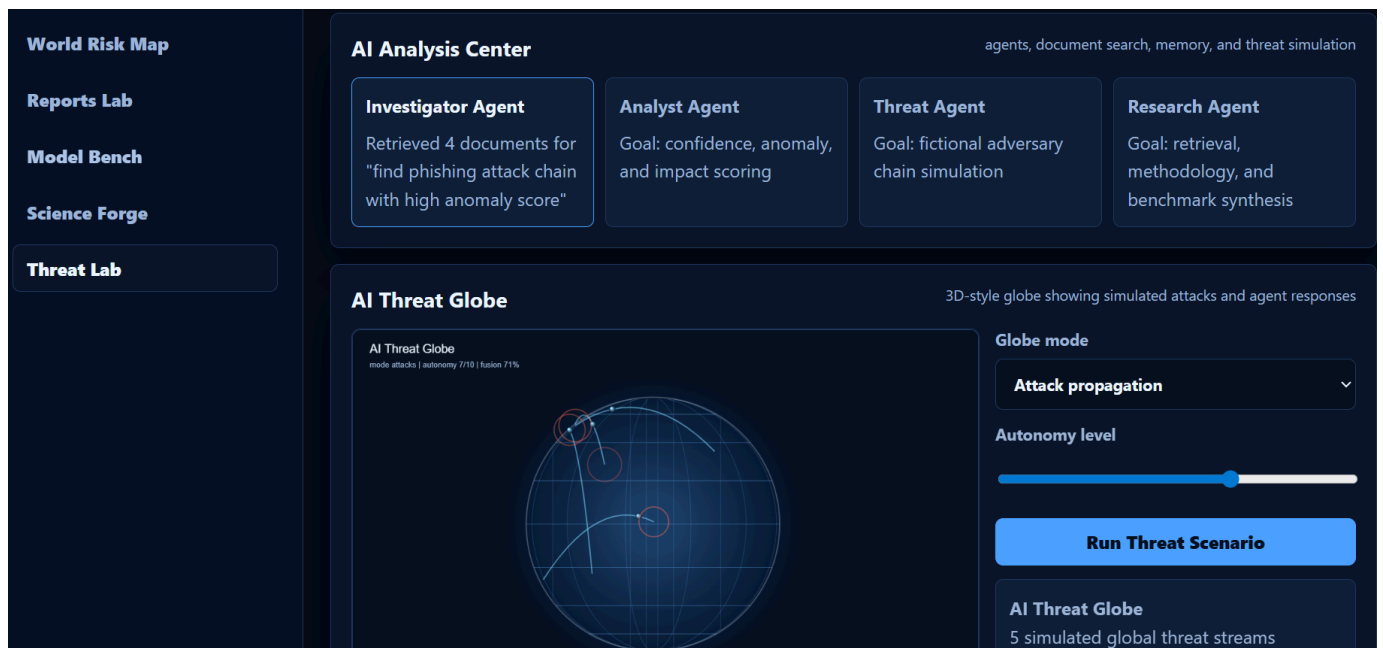
Report lab

Reports Lab is a simulation-based reporting system that generates scientific and intelligence-style assessments from environmental and disaster-response data. The module analyzes risk metrics, exposure levels, and operational readiness to produce structured analytical briefings and automated research reports.



Interactive System Architecture

The Interactive System Architecture represents the core research pipeline powering the platform. It integrates datasets, embedding engines, vector memory systems, simulation modules, and AI agents to enable intelligent analysis, scenario modeling, and automated research synthesis. The architecture was designed to demonstrate scalable browser-based experimentation and multi-agent computational workflows.



Threat Lab & AI Analysis

Threat Lab is an AI-driven cybersecurity simulation platform built for threat analysis, document retrieval, and adversarial scenario modeling. The system integrates multiple analytical agents with an interactive 3D threat visualization engine to simulate attack propagation and autonomous response behavior.

The project demonstrates applied work in AI-agent systems, simulation architecture, and browser-based analytical tooling.

